[~~21MADIP01~~] 22MADIP01

THIRD SEMESTER B. TECH EXAMINATION, DECEMBER – 2023
FOUNDATION OF MATHEMATICS-I

Time: 3 Hours **Maximum Marks: 50**

Sl.no	Questions	A	B	C	D	Correct answer
1	If y=u/vare the functions of x then dy/dx=.....	dy/dx=(v du/dx+u dv/dx)/u^2	dy/dx=(v du/dx-u dv/dx)/u^2	dy/dx=(v du/dx+u dv/dx)/v^2	dy/dx=(v du/dx-u dv/dx)/v^2	
2	If y=cos(ax) then dy/dx=.....	a tan(ax)	- a sin (ax)	- a cos(ax)	a cot(ax)	
3	If y=tan(ax) then dy/dx=.....	a cot(ax)	- a cot(ax)	a (sec)^2(ax)	None of these	
4	If y=cot(ax) then dy/dx=.....	a tan(ax)	a sec(ax)	- a cosec^2(ax)	None of these	
5	If y=sec(ax) then dy/dx=.....	a sec(ax)	a sec(ax) tanx	a tan(ax)	None of these	
6	If y=cosec(ax) then dy/dx=.....	sec(ax)	a sec(ax) cot(ax)	- a coes(ax) cot(ax)	None of these	
7	If 2x^2-3xy+4y^2=1 then dy/dx=.....	(3y-4x)/(8y-3x)	(3y+4x)/(8y+3x)	(4y-3x)/(3y+8x)	None of these	
8	(lim)(x→0)(tanx/x) is.....	1	2	3	4	
9	(lim)(x→0)(x/sinx) is.....	1	0	2	None of these	

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10	If y=e^(-x) then dy/dx=.....	- e^(-x)	- 1/e^x	- x	- 1/x	
11	Find the first order partial derivative with respect to x of f(x,y) = x^2 + y^2	2x	2y	2x+2y	x-y	
12	Find the first order partial derivative with respect to y of f(x,y) = x^3+y^3+3xy	3(y^2)+3x	3(y^2)-3x	(y^2)+x	(y^2)-x	
13	If f(x, y) = sin(y + yx ^2) / 1 + x ^2 Value of f_xy at (0,1) is	0	1	67	90	
14	If u=10xy+6y ^2 then u_x at (1,1) is	12	10	-10	-12	
15	If u=24x^2+6xy+10y , then u_xx is	48	-48	24	-24	
16	Find the first order partial derivative with respect to y of the function f(x,y)=x^3 y^2+3xe^y	6x^2 y+3x	2x^3 y+3xe^y	-6x^2 y+3x	None of these	
17	Find the first order partial derivative with respect to y of the function f(x,y)=x^3 y^2+3xe^y	6x^2 y+3x	2x^3 y+3xe^y	-6x^2 y+3x	None of these	
18	If z= (x^3+y^3)/(3x+4y) then z is a homogeneous function of degree ...?	3	2	1	4	

Sl.no	Questions	A	B	C	D	Correct answer
19	If u= sin^(-1)(x/y)+tan^(-1)(x/y)then x ∂u/∂x+y∂u/∂y=....?	0	2	1	4	
20	If sinu=(x^2 y^2)/(x+y) then x∂u/∂x+y∂u/∂y=....?	3 sinu	3 cosu	3 tanu	None of these	
21	The valve of Γ(1) is equal to	1	2	3	4	
22	The valve of Γ(9/2) is equal to	105/16 √π	106/16 √π	107/16 √π	108/16 √π	
23	∫cosec^2 xdx is equal to	cose cx+c	cotx+c	log(sinx)+c	-cotx+c	
24	∫sin ^(2m-1)θ cos ^ (2n-1)θ dθ From 0 to π/2 =	β(m)	β(m+n)	β(n)	β(m,n)	
25	2∫e^ (-t^2)^ t^(2n-1) dt From 0 to ∞ is equal to	Γ(n)	Γ(n+1)	nΓ(n)	Γ(n+2)	
26	If n is positive integer then Γ(n+1) is equal to	n!	(n+1)!	(n-1)!	None of these	
27	The value of β(m+1,n)+β(m,n+1) is equal to	β(m)	β(m,n)	β(m+n)	β(m-n)	
28	The value of definite integral ∫sin ^7x cos ^5x dx is from 0 to π/2 is	1/160	1/120	3/150	π/160	

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29	The value of β(2,1)+β(1,2) is equal to	1	2	3	4	
30	β(m,n) is equal to	(Γ(m).Γ(n))/(Γ(m-n))	(Γ(m)+Γ(n))/(Γ(m+n))	(Γ(m).Γ(n))/(Γ(m+n))	(Γ(m).Γ(n))/(Γ(m))	
31	∂M/∂x is partial differentiation taking _____ as constant	y as constant	x as constant	Both x and y as constant	None of these	
32	∂M/∂y & ∂N/∂x for the differential equation ydx+(x+y)dy=0 is.....	5	1	0	None of these	
33	∂M/∂y is partial differentiation taking _____ as constant	y as constant	x as constant	Both x and y as constant	None of these	
34	∂M/∂y=∂N/∂x is _____ condition that Mdx+Ndy = 0 to be exact.	Necessary	Sufficient	Both A and B	None of these	
35	A differential equation is said to be exact, if it can be derived from its primitive by_____	Direct Differentiation	Indirect Differentiation	Partial Differentiation	Total Differentiation	
36	Differential equation of family of circles having their centres on the y-axies (x^2+(y-b)^2=r^2) is....	x (d^2 y)/(dx^2)-(dy/dx)^3-dy/dx=0	x^6 (d^2 y)/(dx^2)+(dy/dx)^3-dy/dx=0	(d^2 y)/(dx^2)-(dy/dx) dy/dx=0	None of These	
37	Differential equation of function y=ax^2+bx+c is....	dy/dx=0	(d^2 y)/(dx^2)=0	(d^3 y)/(dx^3)=0	None of these	

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38	Differential equation of the coaxial circles of the system x^2+y^2+2ax+c^2= 0 where c is a constant and a is a variable is....	2xy dy/dx=y^2+x^2+c^2	dy/dx=y^2-x^2	2xy=y^2-x^2+c^2	None of These	
39	Differential equation of the function x^2=4ay is....	dy/dx=(2y^4)/x	dy/dx+2y/5=0	dy/dx=2y/x	None of These	
40	Differential equation of the functionx^3-3x^2 y=c is....	x dy/dx+2y=c	x dy/dx+2y-x=0	dy/dx-x=0	None of these	
41	Nature of the roots 2 , 2 is	Complex	Real and Equal	Real and Distinct	None of these	
42	..... is a particular case of Legendre's linear equation	Linear equation	Cauchy's linear equation	Homogenous equation	None of these	
43	1/D (cosx) is	-sinx	tanx	sinx	None of these	
44	1/D, 1/D^2 , 1/D^3 etc. stands for	Successive differentiation	Successive integration	Integration	Differentiation	
45	1/D^2 (e^3x) is	e^(-3x)/9	e^3x/3	e^(-3x)/3	e^3x/9	
46	A differential equation is considered to be ordinary if it has.....	one dependent variable	more than one dependent variable	one independent variable	more than one independent variable	

Sl.no	Questions	A	B	C	D	Correct answer
47	A pair of linear equations to determine value of two variables is called.....	Simultaneous linear equations	Paired equations	Quadratic equations	Simple equations	
48	An equation of the form a0 (ax+b)^2 y''+a1 (ax+b)y'+a2 y= φ(x) is called	Cauchy's equation	Legendre's equation	Linear equation	None of these.	
49	An equation of the form a0 x^2 y''+a1 xy'+a2 y= φ(x) is called	Linear equation	Legendre's equation	Cauchy's equation	None of these.	
50	By solving the higher order linear differential equation with constant coefficients y=....	y=yp	y=yc	y=yc+yp	y=yc- yp	